

# Dividing by a One-Digit Number

## Answer Key

### Grade 4 — Whole-Number Division

#### Quick Answers

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| 1. 21                                                                                                                                   | 7. (a) the quotient = 103, —                                                                                        |
| 2. 16                                                                                                                                   | 8. quotient = 578, remainder = 5                                                                                    |
| 3. $6 \times 40 = 240$ , what is left after 40 groups = 18, groups in what is left = 3, total quotient = 43                             | 9. (a) 4825 rounded to the nearest thousand = 5000, (b) the estimated quotient = 1000, (c) the exact quotient = 965 |
| 4. quotient = 59, remainder = 3                                                                                                         | 10. $7 \times 600 = 4200$ , what is left after 600 groups = 672, groups in what is left = 96, total quotient = 696  |
| 5. (a) 3142 rounded to the nearest thousand = 3000, (b) the estimated quotient = 500, (c) exact quotient = 523, (c) exact remainder = 4 | 11. 9                                                                                                               |
| 6. 321                                                                                                                                  | 12. (a) the quotient = 1809, —                                                                                      |
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#### What is verified

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25 of 27 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 7, 12. The worked solution states what a correct response must contain.

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#### Curriculum

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**Level:** Grade 4 — Whole-Number Division

4.NBT.B.6 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

*Difficulty 1–5 of 5 across 27 tagged checks.*

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#### Common wrong answers

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7. *If they got 13: left the zero out of the tens place and wrote a two-digit quotient*

12. *If they got 189: dropped the zero from the tens place of the quotient and wrote a three-digit answer*

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1. Divide  $84 \div 4$ .

**Solution:** 8 tens  $\div 4 = 2$  tens; 4 ones  $\div 4 = 1$  one. Check:  $21 \times 4 = 84$ .

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| 21 |
|----|

2. Divide  $96 \div 6$ .

**Solution:** 6 goes into 9 once with 3 left, so write 1 in the tens place and carry the 3 tens down:  $36 \div 6 = 6$ . Check:  $16 \times 6 = 96$ . 16

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3. Divide  $258 \div 6$  with partial quotients.

**Solution:** Take out an easy chunk first:

$$6 \times 40 = \boxed{240}, \quad 258 - 240 = \boxed{18}$$

Now finish the leftover:  $18 \div 6 = \boxed{3}$ . Add the partial quotients:  $40 + 3 = 43$ . Check:  $43 \times 6 = 258$ . 43

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4. Divide  $475 \div 8$ .

**Solution:** 8 does not go into 4, so start with 47:  $8 \times 5 = 40$ , leaving 7; bring down the 5 to make 75;  $8 \times 9 = 72$ , leaving 3. The remainder 3 is smaller than 8, so the division is finished. Check:  $59 \times 8 + 3 = 472 + 3 = 475$ . 59

remainder

3

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5.  $3,142 \div 6$ : (a) round, (b) estimate, (c) exact.

**Solution (a):** The hundreds digit of 3142 is 1, below 5, so it rounds down. 3000

**Solution (b):**  $3000 \div 6 = 500$ , so expect a quotient a little over five hundred. 500

**Solution (c):** 6 into 31 goes 5 times (30), leaving 1; bring down 4 to make 14; 6 goes twice (12), leaving 2; bring down 2 to make 22; 6 goes 3 times (18), leaving 4. The estimate of 500 sits just below the exact quotient, as expected after rounding down. 523

remainder

4

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6. Divide  $2,568 \div 8$ .

**Solution:** 8 into 25 goes 3 times (24), leaving 1; bring down 6 to make 16; 8 goes twice, leaving 0; bring down 8; 8 goes once. Check:  $321 \times 8 = 2568$ . 321

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7. Divide  $618 \div 6$ . (a) The quotient. (b) Why must the zero be written?

**Solution (a):** 6 into 6 goes once, leaving 0; bring down 1 — 6 does not go into 1, so the tens digit of the quotient is 0; bring down 8 to make 18, and 6 goes 3 times. Check:  $103 \times 6 = 618$ . 103

**Solution (b) — instructor-judged.** A full-credit answer explains that there is no whole ten of groups left after the hundred, so a 0 has to hold the tens place and keep the 1 in the hundreds column and the 3 in the ones column. Model answer: “6 won’t go into 1, so I write 0 above that digit. Without it the answer would read 13, which is nowhere near  $618 \div 6$ .” Giving only the quotient, or saying the zero appears because a digit was skipped, is not full credit.

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8. Divide  $5,207 \div 9$ .

**Solution:** 9 into 52 goes 5 times (45), leaving 7; bring down 0 to make 70; 9 goes 7 times (63), leaving 7; bring down 7 to make 77; 9 goes 8 times (72), leaving 5. Check:  $578 \times 9 + 5 = 5202 + 5 = 5207$ .

remainder

578

5

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- 9.**  $4,825 \div 5$ : (a) round, (b) estimate, (c) exact.

**Solution (a):** The hundreds digit of 4825 is 8, which is 5 or more, so it rounds up.

5000

**Solution (b):**  $5000 \div 5 = 1000$ .

1000

**Solution (c):** 5 into 48 goes 9 times (45), leaving 3; bring down 2 to make 32; 5 goes 6 times (30), leaving 2; bring down 5 to make 25; 5 goes exactly 5 times. No remainder. The exact quotient is just under the estimate of 1000, which is right because the dividend was rounded up.

965

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- 10.** Divide  $4,872 \div 7$  with partial quotients.

**Solution:** Take out a large easy chunk first:

$$7 \times 600 = \boxed{4200}, \quad 4872 - 4200 = \boxed{672}$$

Then finish the leftover:  $672 \div 7 = \boxed{96}$  (because  $7 \times 90 = 630$  and  $7 \times 6 = 42$ ). Add the partial quotients:  $600 + 96 = 696$ . Check:  $696 \times 7 = 4872$ .

696

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- 11.** 1,908 divided by a one-digit number gives exactly 212. What is the divisor?

**Solution:** This is division read backwards. If  $1908 \div \square = 212$  then  $212 \times \square = 1908$ , so the divisor is  $1908 \div 212$ . Estimating:  $212 \times 9 = 1908$  exactly, while  $212 \times 8 = 1696$  is too small.

9

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- 12.** Divide  $7,236 \div 4$ . (a) The quotient. (b) How could you tell in advance it has four digits?

**Solution (a):** 4 into 7 goes once (4), leaving 3; bring down 2 to make 32; 4 goes 8 times, leaving 0; bring down 3 — 4 does not go into 3, so write 0; bring down 6 to make 36, and 4 goes 9 times. Check:  $1809 \times 4 = 7236$ .

1809

**Solution (b) — instructor-judged.** A full-credit answer gives a reason available before dividing. Model answer 1: “4 goes into the leading 7 at least once, so the first digit of the quotient sits above the thousands place, and a quotient that starts in the thousands has four digits.” Model answer 2: “About 7000 shared four ways is somewhere between 1000 and 2000, so the answer has four digits.” Simply stating 1809 is not full credit — the question asks for the reasoning that comes first.